

EUREKA, NU, Canada

WMO: 719170

Lat: 79.989N

Lon: 85.934W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-47.8	-46.0	-48.2	0.0	-43.7	-46.7	0.0	-43.4	15.6	-23.6	12.3	-27.0	0.5	130	1.145

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	13.2	7.9	11.3	7.0	9.7	6.1	8.2	12.6	7.2	11.0	6.3	9.4	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.0	5.4	8.7	4.0	5.0	8.5	3.1	4.7	7.9	25.2	12.8	23.2	11.1	21.2	9.5	12.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.0	8.6	DB	-48.7	16.2	1.7	2.9	-49.9	18.3	-50.9	20.0	-51.9	21.6	-53.1	23.7	(4)
(5)			WB	-46.6	9.8	1.4	1.5	-47.6	10.9	-48.4	11.8	-49.1	12.6	-50.1	13.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-17.9	-35.5	-36.7	-35.9	-25.8	-9.7	3.1	6.6	3.6	-5.5	-19.5	-27.7	-32.8		
	DBStd	17.07	6.00	6.13	6.86	6.59	5.90	3.39	2.78	3.19	4.94	7.33	6.78	7.00		
	HDD10.0	10178	1410	1309	1421	1075	610	207	111	201	464	913	1132	1326		
	HDD18.3	13213	1668	1542	1679	1325	868	456	363	458	714	1172	1382	1584		
	CDD10.0	7	0	0	0	0	0	1	6	1	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.0	2.2	2.1	1.9	2.2	3.3	4.8	4.8	4.3	3.5	2.2	2.1	2.2		
(15) Precipitation	PrecAvg	75	3	3	3	4	3	8	13	14	10	8	4	3		
	PrecMax	141	11	15	12	21	15	31	46	37	28	26	18	24		
	PrecMin	38	0	0	0	0	0	0	0	1	1	1	0	0		
	PrecStd	21	2	3	3	4	3	6	11	9	7	5	4	4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-15.9	-16.4	-12.9	-6.1	3.2	14.0	17.5	12.8	3.7	-1.7	-7.5	-9.0		
		MCWB	-16.7	-17.0	-13.4	-7.0	0.8	8.1	9.8	7.9	1.5	-2.5	-8.4	-9.5		
	2%	DB	-20.2	-20.9	-17.6	-9.8	1.5	11.0	14.7	10.7	2.3	-3.9	-10.7	-14.5		
		MCWB	-20.7	-21.3	-18.1	-10.6	-0.3	6.3	8.6	6.9	0.8	-4.6	-11.4	-15.0		
	5%	DB	-23.4	-24.2	-22.2	-13.9	-0.4	9.1	12.7	9.0	1.4	-6.1	-14.0	-18.4		
		MCWB	-23.8	-24.7	-22.6	-14.6	-1.8	5.3	7.8	6.1	0.0	-6.7	-14.6	-18.9		
	10%	DB	-26.5	-27.5	-25.5	-16.7	-2.1	7.5	11.1	7.7	0.4	-8.7	-17.4	-21.7		
		MCWB	-26.8	-27.7	-25.8	-17.3	-3.3	4.4	6.9	5.2	-0.8	-9.2	-18.0	-22.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.5	-16.8	-13.2	-7.0	1.1	8.2	10.2	8.2	2.0	-2.4	-8.5	-9.6		
		MCDB	-15.9	-16.1	-12.7	-6.1	2.8	13.5	16.5	11.7	3.3	-1.8	-7.7	-9.0		
	2%	WB	-20.5	-21.1	-18.1	-10.6	-0.2	6.5	8.9	7.3	1.1	-4.5	-11.3	-15.1		
		MCDB	-20.0	-20.8	-17.6	-9.8	1.5	10.6	14.0	10.2	2.1	-3.9	-10.7	-14.5		
	5%	WB	-23.7	-24.3	-22.4	-14.6	-1.7	5.4	7.9	6.3	0.2	-6.7	-14.6	-18.9		
		MCDB	-23.3	-23.9	-22.0	-13.9	-0.4	8.9	12.3	8.7	1.3	-6.2	-14.0	-18.4		
	10%	WB	-26.6	-27.3	-25.5	-17.3	-3.3	4.4	7.0	5.4	-0.8	-9.2	-18.0	-22.1		
		MCDB	-26.3	-27.1	-25.2	-16.7	-2.0	7.3	10.8	7.6	0.3	-8.6	-17.4	-21.8		
(35) Mean Daily Temperature Range	MDBR		6.1	6.3	5.9	6.7	5.3	4.3	5.4	3.8	4.0	6.1	6.2	6.0		
	5% DB	MCDBR	9.9	10.2	9.5	7.8	5.0	7.1	8.3	5.8	3.4	6.0	8.8	9.2		
		MCWBR	9.7	10.0	9.4	7.4	4.0	4.3	4.7	3.5	2.7	5.6	8.5	8.9		
	5% WB	MCDBR	9.9	10.1	9.7	7.7	5.0	6.9	7.9	5.6	3.2	5.9	8.9	9.0		
		MCWBR	9.7	9.9	9.5	7.4	4.1	4.3	4.7	3.6	2.6	5.6	8.6	8.8		
(40) Clear-Sky Solar Irradiance	taub		N/A	N/A	0.208	0.254	0.285	0.318	0.320	0.296	0.216	N/A	N/A	N/A		
	taud		N/A	N/A	2.053	2.206	2.231	2.359	2.465	2.466	2.266	N/A	N/A	N/A		
	Ebn at Noon		0	0	564	784	828	815	798	746	597	0	0	0		
	Edh at Noon		0	0	41	74	92	87	75	60	35	0	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.00	0.70	3.57	6.57	6.72	5.61	3.27	1.03	0.00	0.00	0.00		
	RadStd		0.00	0.00	0.08	0.21	0.31	0.56	0.52	0.33	0.08	0.00	0.00	0.00		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page